

LAB 3 MOSFET I-V Characteristics

Technology: 22nm PTM-HP

Nominal VDD = 0.8 V

$k = 1.3$ (W_p/W_n), $W_n = 44\text{nm}$, $W_p = k \cdot W_n$ (from Lab 1)

Single PMOS (MP1) and single NMOS (MN1) devices, each characterized separately with VGATE and VDRAIN sweeping the gate and drain nodes. I_D is reported in a consistent drain-to-source reference direction for both devices.

Lab 3.0

PMOS Drain current vs. VDS, family of curves over VGS

RON (from plot) = 9.09 kohm

RON (calculated) = 3.80 kohm = $(1/(u_0 C_{ox}(W/L) \cdot V_{ov}))$,

$u_0 = 0.0095 \text{ m}^2/\text{Vs}$, $C_{ox} = 3.14 \times 10^{-2} \text{ F/m}^2$ from $t_{oxe} = 1.1\text{nm}$, $W/L = 2.6$, $V_{ov} = 0.8 - 0.4606 = 0.339\text{V}$)

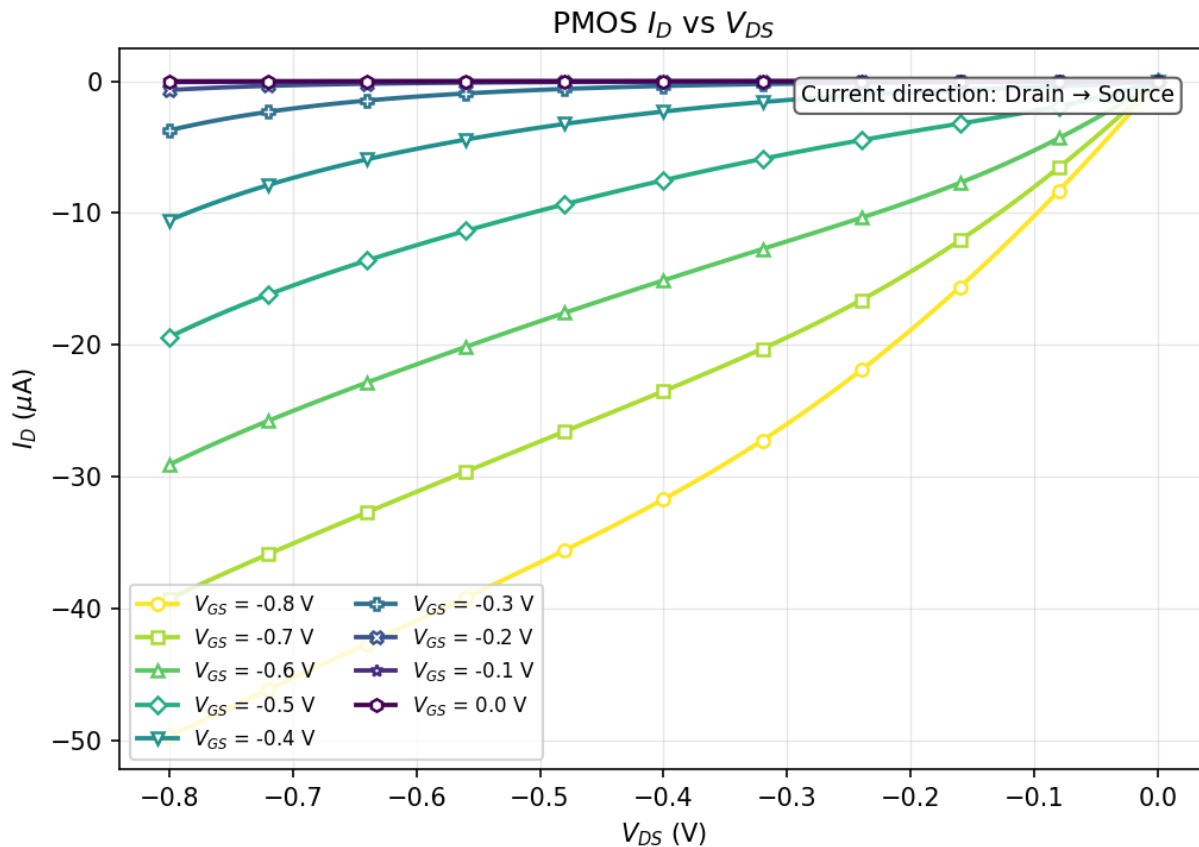


Figure 1: PMOS I_D vs V_{DS} family of curves

Lab 3.1

NMOS drain current vs. VDS, family of curves over VGS

RON (from plot) = 5.55 kohm RON (calculated) = 1.28 kohm = $(1/(u_0 C_{ox}(W/L) \cdot V_{ov}))$,

$u_0 = 0.04 \text{ m}^2/\text{Vs}$, $C_{ox} = 3.29 \times 10^{-2} \text{ F/m}^2$ from $t_{oxe} = 1.05\text{nm}$, $W/L = 2.0$, $V_{ov} = 0.8 - 0.50308 = 0.297\text{V}$)

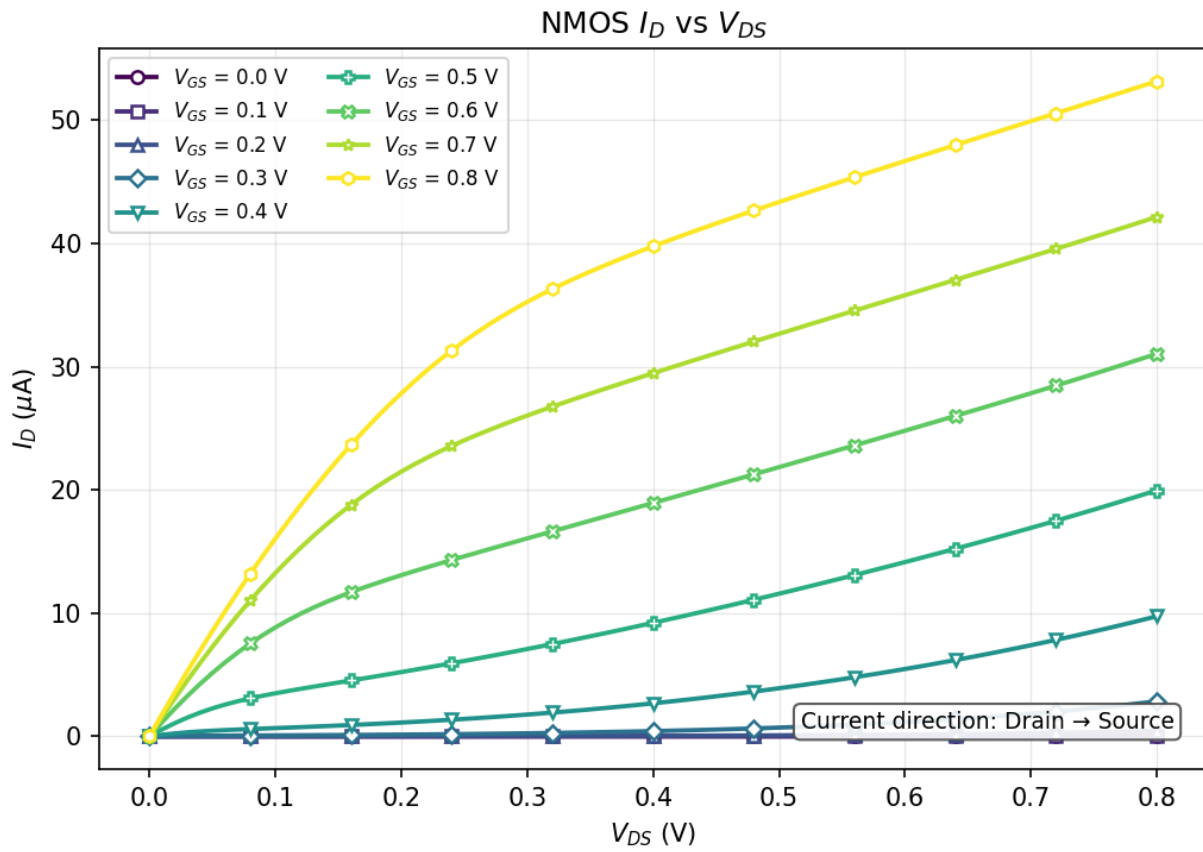


Figure 2: NMOS I_D vs V_{DS} family of curves

Appendix: I-V sweep data

Full per-point sweep data (VGS, VDS, ID) for both devices is in CSV form:

- PMOS: `results_pmos/iv_table.csv`
- NMOS: `results_nmos/iv_table.csv`